

ANTING MA

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OBJECTIVE

Self-motivated Computer Science graduate with hands-on experience as a Data Engineer Apprentice at Pfizer, building scalable data pipelines and relational database solutions. Proficient in Python, SQL, ETL processes, and Agile methodologies, with experience automating HPC workflows and developing interactive visualizations. Eligible to work in the U.S. without sponsorship. Eager to apply technical skills to drive efficient data management and analysis.

EDUCATION

University of Connecticut, Storrs, CT

May 2024

Bachelor of Science, Major in Computer Science, Minor in Mathematics

Concentration: Software Design and Development

EXPERIENCE

Pfizer

Sep 2024 - Feb 2025

Data Engineer Apprentice

- Established a modular **Python-based pipeline** using **Pandas** and **RDKit** to compute Lipinski properties for **large-scale compound datasets** (100M+ records), filter results, and export **SMILES, JSON, and XML** outputs.
- Deployed and **queried** a **SQLite** database to store compound descriptors, rule-level pass/fail flags, and batch metadata; applied **constraints** and **SMILES hashing** for efficient deduplication and incremental loading.
- Automated **pipeline** execution on an **HPC cluster** using **Slurm** and **shell scripts**, and generated insightful **visualizations** with **Plotly** to summarize per-rule pass/fail statistics and overall compound quality.
- Engineered scalable **data pipelines** to **extract, transform, and aggregate** structured data from **18,000+** compressed experiment files; enabled dataset labeling and feature generation for machine learning workflows.
- Built interactive **visualization** tools using **Plotly** to support threshold selection; automated dataset preparation and model training with performance **evaluation metrics** (accuracy, kappa, confusion matrix).
- Created an easy-to-use **interactive visualization tool** in Jupyter notebook for adventitious virus detection in biologics manufacturing, supporting **data-driven insights for quality control**.

Boehringer Ingelheim

Sep 2022 - Apr 2023

Target Annotation (Project Contributor)

- Designed and implemented a relational **PostgreSQL** database to store structured **UniProt** data; defined efficient schema, indexed key fields, and wrote **SQL queries** to optimize data retrieval and ensure data integrity.
- Constructed a scalable **ETL pipeline** using **Python** and **Pandas** to retrieve data from UniProt's **REST API**, parse **JSON** responses, normalize fields, and populate relational tables in PostgreSQL for efficient querying.
- Built a **Flask** web interface integrated with **SQL queries** to allow users to explore protein data, rate research articles, and leave comments, supporting collaborative scientific analysis and reproducible workflows.

Breakthrough Tech AI - MIT

Sep 2022 - Dec 2022

Machine Learning Assisted Whiteboard (Project Contributor)

- Developed a user-friendly frontend using **Next.js, React, and TailwindCSS** to enable non-technical users to generate images from custom prompts using ML models like Stable Diffusion.
- Handled prompt processing and AI image generation with **JavaScript**, supporting responsive layouts, annotation tools, and seamless downloads to enhance accessibility and user engagement.

SKILLS

Certification: AWS Cloud Practitioner, Certificate in Machine Learning Foundations (eCornell)

Programming: Python, Bash (Shell) Scripting, C, C++, Scheme, Java, Kotlin, RISC-V, JavaScript

Database Management: SQL, NoSQL, PostgreSQL, MySQL, SQLite, MongoDB

Data Science and Machine Learning: Pandas, Polars, Scikit-Learn, Numpy, TensorFlow

Data Visualization: Matplotlib, Plotly, Seaborn, Tableau

Web Development: HTML, CSS, Next.js, React, REST APIs, Flask, FastAPI, TailwindCSS

Tools & Platforms: Linux, Git, GitHub, Docker